



Development and Validation of a Risk Calculator Predicting Postoperative Respiratory Failure

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Background: Postoperative respiratory failure (PRF) (requiring mechanical ventilation > 48 h after surgery or unplanned intubation within 30 days of surgery) is associated with significant morbidity and mortality. The objective of this study was to identify preoperative factors associated with an increased risk of PRF and subsequently develop and validate a risk calculator.

Methods: The American College of Surgeons National Surgical Quality Improvement Program (NSQIP), a multicenter, prospective data set (2007-2008), was used. The 2007 data set (n = 211,410) served as the training set and the 2008 data set (n = 257,385) as the validation set.

Results: In the training set, 6,531 patients (3.1%) developed PRF. Patients who developed PRF had a significantly higher 30-day mortality (25.62% vs 0.98%, $P < .0001$). On multivariate logistic regression analysis, five preoperative predictors of PRF were identified: type of surgery, emergency case, dependent functional status, preoperative sepsis, and higher American Society of Anesthesiologists (ASA) class. The risk model based on the training data set was subsequently validated on the validation data set. The model performance was very similar between the training and the validation data sets (c-statistic, 0.894 and 0.897, respectively). The high c-statistics (area under the receiver operating characteristic curve) indicate excellent predictive performance. The risk model was used to develop an interactive risk calculator.

Conclusions: Preoperative variables associated with increased risk of PRF include type of surgery, emergency case, dependent functional status, sepsis, and higher ASA class. The validated risk calculator provides a risk estimate of PRF and is anticipated to aid in surgical decision making and informed patient consent.

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Abbreviations: ACS = American College of Surgeons; ASA = American Society of Anesthesiologists; BIC = Bayes Information Criterion; NSQIP = National Surgical Quality Improvement Program; PRF = postoperative respiratory failure; ROC = receiver operating characteristic; SCNR = surgical clinical nurse reviewer; VA = Veterans Affairs; VASQIP = Veterans Affairs Surgical Quality Improvement Program

The benefits of any surgical procedure are heavily influenced by the accompanying morbidity and mortality. Complications after surgery not only worsen outcomes but also prolong hospital stay and are associated with a significantly increased cost in hospital

care.¹⁻³ Pulmonary complications account for 10% to 40% of postoperative complications after abdominal and vascular surgeries.^{4,6}

Postoperative respiratory failure (PRF) is commonly understood as failure to wean from mechanical ventilation within 48 h of surgery or unplanned

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intubation/reintubation postoperatively. It is one of the most serious pulmonary complications and is associated with marked increases in length of stay, morbidity, and mortality.^{7,8} Most previous studies assessing risk factors for postoperative pulmonary complications have studied them in aggregate; few studies have assessed risk factors associated solely with PRF. As most of these are single-institution, retrospective studies, their limitations are extensive.^{9,10} Although two large multicenter studies have assessed the risk factors associated with PRF, both were Veterans Affairs Surgical Quality Improvement Program (VASQIP) based.^{7,8} One included only patients from Veterans Affairs (VA) and excluded women. The other pooled patients from 128 VA hospitals with those from 14 non-VA hospitals, again making their population primarily VA-based. Further, although one of these studies divided surgeries based on the incision site—and not the organ involved—the other included only general and vascular surgery cases.

Since the publication of these two multicenter studies, there has been significant evolution of the National Surgical Quality Improvement Program (NSQIP) data set. The sample size has grown significantly (> 180 hospitals now contribute data), and data for women and cardiac surgery patients are now included.¹¹ The database is thus not only useful for institutional and individual quality assessment, but it provides a mechanism to facilitate further improvement in outcomes.

Given the paucity of comprehensive data on the subject, we used the American College of Surgeons (ACS) NSQIP database to study the association of PRF with postoperative length of stay, morbidity, mortality, and other clinical outcomes. We analyzed the NSQIP data set to assess the risk factors for PRF among all surgical patients, including a comparison of different types of surgeries with respect to their risk of developing PRF. We used these risk factors to develop a validated risk calculator. Knowledge of these risk factors might help guide optimization of select preoperative medical conditions, which may reduce the incidence of PRF and improve outcomes.

MATERIALS AND METHODS

Data Set

As this study used a publicly available national data set, it was exempt from IRB review. Data were extracted from the 2007 and 2008 ACS NSQIP Participant Use Data Files.¹¹ These are multicenter, prospective databases with 183 (year 2007) and 211 (year 2008) participant academic and community US hospitals. In NSQIP, a participant hospital's surgical clinical nurse reviewer (SCNR) captures data using a variety of methods, including medical chart abstraction. The data are collected based on strict criteria formulated by a committee. To ensure high-quality data

collection, NSQIP has developed different training mechanisms for the SCNR and conducts an interrater reliability audit of participating sites.¹¹ The combined results of the audits revealed an overall disagreement rate of approximately 1.99% for all assessed program variables. The processes of SCNR training, interrater reliability auditing, data collection, and sampling methodology have been previously described in detail.¹¹⁻¹⁴

Patients

Patients who underwent surgeries listed in Table 1 were studied (2007 data set: $n = 211,410$; 2008 data set: $n = 257,385$). Data obtained included demographic, lifestyle, comorbidity, and other variables. The list of variables extracted is mentioned in e-Appendix 1.

Outcome

The primary end point was PRF through 30 days following surgery. PRF was said to have occurred if patients: (1) had an unplanned intubation during their surgery or postoperatively, (2) were reintubated postoperatively once extubated, or (3) required mechanical ventilation for > 48 h postoperatively.

If the patient returned to the operating room for any reason and was intubated as part of the anesthesia/surgery, then it was not counted as a reintubation. If a patient self-extubated and had to be reintubated, then also it was not counted as a reintubation.

Statistical Analysis

Using patients from the 2007 NSQIP data set, univariate exploratory analysis was performed using Pearson χ^2 test or Fisher exact test for categorical variables, and T or F test for continuous variables. Stepwise multivariate logistic regression was carried out to assess risk factors predictive of PRF, thus creating the "full" model. To address the possibility that the surgery type might interact with any of the other variables, we included all second-order interaction terms involving surgery into the list of candidate variables.

To reduce the number of risk factors, we then created a parsimonious model by sequentially removing variables from the full model. A forward selection procedure was applied to select a predetermined number of variables into the final logistic model from a list of candidate variables, which in this analysis includes all preoperative variables. A Bayes Information Criterion (BIC) vs model size plot was created to determine the number of variables in the final parsimonious model.¹⁵ The model with 21 variables gave the lowest BIC, with a minimal increase in BIC on further reduction of the number of variables to five. Thus, because of the minimal difference in BIC and to reduce the complexity and make the model practically usable, the number of variables included in the final parsimonious model was determined as five. Furthermore, more than five variables in the model did not significantly improve the c-statistic, and there was also a minimal loss in the calibration with this approach. Statistical analysis was performed using SAS, version 9.2 (SAS Institute; Cary, North Carolina). P value < .05 was considered significant.

Risk Model Performance

The accuracy of a logistic regression model is usually assessed by its discrimination and calibration, with both being used in this study.¹⁶ Discrimination measures how well a model can distinguish between cases (who develop PRF) vs noncases (who do not develop PRF). Discrimination is usually assessed by c-statistic, also known as the area under the receiver operating characteristic

Table 1—Description of Surgeries

Type of Surgery	Includes	No. (%)
Anorectal	Surgeries involving anus and rectum with transanal approach	3,265 (1.5)
Aortic	Surgeries involving the aorta	4,479 (2.1)
Bariatric	Bariatric surgeries	12,337 (5.8)
Brain	Surgeries involving brain	370 (0.2)
Breast	Surgeries involving breast	21,359 (10.1)
Cardiac	Surgeries involving heart and not aorta	631 (0.3)
ENT	ENT and head and neck surgeries except thyroid and parathyroid	646 (0.3)
GBAAS	Surgeries involving gall bladder, appendix, adrenals or spleen. Biliary tree surgeries other than cholecystectomy not included	35,940 (17.0)
FG/HPB	Foregut and hepatopancreatobiliary surgeries: esophagus, stomach, duodenum, pancreas, liver and biliary tree (except isolated cholecystectomy)	12,254 (5.8)
Hernia	Surgeries involving ventral, inguinal, femoral and other hernias (except hiatal hernias)	31,692 (15.0)
Intestine	Surgeries involving intestines below the level of duodenum using abdominal approach	31,492 (14.9)
Neck	Thyroid and parathyroid surgeries	10,179 (4.8)
Nonesophageal thoracic	Surgeries in the thorax excluding heart and esophagus	1,324 (0.6)
OB/GYN	Obstetric and gynecologic surgeries	2,861 (1.4)
Orthopedic	Surgeries involving orthopedics and nonvascular extremity	9,272 (4.4)
Other abdomen	Abdominal surgeries not covered by above	4,086 (1.9)
Peripheral vascular	Nonaortic, nonvein vascular surgeries	17,490 (8.3)
Skin	Surgeries involving skin	4,906 (2.3)
Spine	Surgeries involving spine	1,469 (0.7)
Urology	Surgeries involving kidneys and urinary system	1,975 (0.9)
Vein	Surgeries involving just veins	3,383 (1.6)

N = 211,410. ENT = ear, nose, throat; FG = foregut; GBAAS = gall bladder, appendix, adrenals, spleen; GYN = gynecologic; HPB = hepatopancreatobiliary; OB = obstetric.

(ROC) curve. The c-statistic ranges from 0.50 (no better than flipping a coin) to 1.00 (model is 100% correct).

Calibration (Hosmer-Lemeshow test) measures a model's ability to generate predictions that are on average close to the average observed outcome. In studies with large sample sizes, it is suggested to construct a calibration graph of observed vs predicted event.¹⁶ If the model calibrates well, there will not be a substantial deviation from the 45-degree line of perfect fit.

Risk Model Validation

Once a suitable model was chosen based on the 2007 data set, an independent data set (2008 data set) was used to validate the model. The model validation applied the trained model from the 2007 data set to estimate PRF probabilities for all patients in the 2008 data set. These estimated probabilities were then compared with actual PRF status in the 2008 data set by computing a c-statistic. To do this, an ROC curve was constructed based on the sensitivity and specificity of the predictions from the 2007 model on the 2008 data set for various prediction cut points. The c-statistic is equivalent to the area under this ROC curve and was computed using the trapezoidal rule. This c-statistic reflects how much predictive accuracy the trained (2007) model has on the 2008 data set. If this c-statistic shows favorable predictive accuracy, then the model is considered validated. As previously described in literature, similar results for discrimination indicate validation in an independent data set.^{7,17-19}

Development of Risk Calculator

Once the model was validated, it was used to develop the risk calculator, which takes the form of an interactive spreadsheet that accepts patient covariate information and returns estimated probability percentage of PRF based on the validated model. Alternatively, one can generate the estimated PRF probability

directly using the logistic regression model. The multiple logistic regression model is as follows:

$$L = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \dots + \beta_k x_k$$

where L represents the natural log of the odds of PRF, β_0 is the intercept for the model, k represents the number of parameters needed for the preoperative predictors of PRF, β_1 through β_k represent the model parameters corresponding to the selected predictors of PRF, and x_1 through x_k represent patient data corresponding to the selected predictors of PRF. Categorical predictors such as procedure type are incorporated into the model using reference coding. This means that one level of the categorical predictor is chosen as a reference category, and the remaining levels of the predictor are compared with the reference. For example, there are 21 different entries for procedure (Table 1), so the model has 20 β parameters and x variables that describe the difference between the reference procedure (hernia surgery) and each of the other 20 procedures in terms of log-odds of PRF.

The parameter estimates and standard errors for the model are presented in the results section. These estimated coefficients can be used to estimate the logit for a patient by substituting the patient's data along with the estimated coefficients into the logistic regression equation. If the patient belongs to the reference group for a categorical variable, then all of the x values associated with that variable in the model are zero. If the patient belongs to a nonreference group, then the appropriate x value is equal to one, and all other x values associated with the categorical variable are zero. The estimated probability percentage of PRF for a patient is then computed using the following formula:

$$\text{estimated probability} = \frac{e^L}{1 + e^L}$$

Table 2—Univariate Analysis—Preoperative (2007 NSQIP Data Set)

Variables	PRF (n = 6,531)	No PRF (n = 204,879)
Acute renal failure with rising creatinine > 3	6.60	0.42
Age	65.9 ± 14.8	54.9 ± 17.2
Angina within 1 mo	3.28	0.88
ASA class		
1	0.3	10.8
2	8.2	46.4
3	43.3	37.0
4	43.5	5.7
5	4.7	0.2
Ascites	12.46	1.41
Bleeding disorder	21.13	5.71
BMI	30.1 ± 9.4	30.7 ± 8.8
Chemotherapy within 30 d	2.16	1.13
CHF within 1 mo	8.54	0.80
Coma > 24 h	1.26	0.04
COPD with FEV ₁ < 75% or causing functional disability or hospitalization	17.32	4.34
Corticosteroid use	9.31	3.19
Diabetes	26.40	14.18
On dialysis	8.42	2.21
Disseminated cancer	4.98	1.94
DNR	2.50	0.56
Emergency case	44.37	11.58
Esophageal varices on EGD/CT scan in last 6 mo	0.75	0.12
ETOH > 2 drinks/d within 2 wk of surgery	5.36	2.43
Functional status (independent)	53.48	94.31
Functional status (partially independent)	19.14	4.39
Functional status (totally dependent)	27.38	1.30
Hemiplegia	3.17	0.93
Hypertension requiring medications	69.15	44.38
Male gender	53.15	42.49
Myocardial infarction within 6 months	4.59	0.61
Open wound	18.40	4.49
Paraplegia	1.27	0.38
Preexisting pneumonia	7.66	0.34
Previous PCI	11.42	5.21
Previous cardiac surgery	14.99	5.84
Prior operation within 30 d	18.08	3.09
PVD with revascularization/amputation	9.66	4.24
Quadriplegia	0.69	0.11
Race (black)	12.42	9.76
Radiotherapy within 90 d	1.58	0.75
Rest pain in lower extremity due to PVD	5.07	2.45
Sepsis		
None	52.58	92.14
SIRS	19.46	0.46
Sepsis	18.65	6.06
Septic shock	9.31	1.34
Smoker within last year	27.81	20.67
Smoker pack-y	23.0 ± 32.7	10.8 ± 22.4
Stroke with neurologic deficit	6.94	2.32
Stroke with no neurologic deficit	4.73	1.92
TIA history	5.22	2.86
Transfusion > 4 units PRBC preoperative	4.92	0.26
Tumor involving CNS	0.51	0.12
Ventilator dependent during last 48 h	20.21	0.41
> 10% weight loss within 6 mo	7.84	2.39
Abnormal laboratory values ^a		
Sodium	31.67	12.94
BUN	61.17	37.78
Creatinine	32.33	8.96
Albumin	66.63	22.14
Alkaline phosphatase	25.01	14.26

(Continued)

Table 2—Continued

Variables	PRF (n = 6,531)	No PRF (n = 204,879)
Bilirubin	30.86	16.96
HCT	56.43	23.39
Platelet	37.87	19.48
PT	63.05	39.22
PTT	31.11	13.17
SGOT	33.38	16.35
WBC	46.86	23.22

All values except continuous variables are expressed as percentages. Continuous variables are expressed as mean $\pm$ SD. All *P* values $\leq .0001$; *P* value for pregnancy = .1026. ASA = American Society of Anesthesiologists; CHF = congestive heart failure; CNS = central nervous system; DNR = do not resuscitate; EGD = esophagogastroduodenoscopy; ETOH = alcohol; HCT = hematocrit; NSQIP = National Surgical Quality Improvement Program; PCI = percutaneous coronary intervention; PRBC = packed RBC; PRF = postoperative respiratory failure; PT = prothrombin time; PTT = partial thromboplastin time; PVD = peripheral vascular disease; SGOT = serum glutamic oxaloacetic transaminase; SIRS = systemic inflammatory response syndrome; TIA = transient ischemic attack.

*Abnormal laboratory values: sodium (<136 or >145 mEq/L), BUN (<10 or >20 mg/dL), creatinine (>1.5 mg/dL), albumin (<3.5 g/dL), alkaline phosphatase (<120 IU/L), bilirubin (>2 mg/dL), hematocrit ($<36\%$), platelets ($<150,000$ or $>350,000$ per μL), PT (>13.5 s), PTT (>35 s), SGOT (>35 IU/L), WBC ($<4,500$ or $>11,000$ cells/mL).

RESULTS

Univariate Analysis (2007 Data Set)

Of 211,410 patients in the 2007 NSQIP data set, PRF was seen in 6,531 patients (3.1%). In the 2008 data set used for validation (n = 257,385), PRF was seen in 6,590 patients (2.6%).

PRF was significantly associated with a multitude of variables ($P < .0001$ for all) (Tables 2, 3). Postoperatively, patients with PRF had more complications than those without PRF. Death within 30 days was significantly higher in patients with PRF (25.62% vs 0.98%; $P < .0001$).

Multivariate Analysis for PRF (2007 Data Set)

Preoperative variables significantly associated with an increased risk for PRF in the 21-variable model are mentioned in e-Appendix 2. Preoperative variables significantly associated with an increased risk for PRF in the final model included American Society of Anesthesiologists (ASA) class, dependent functional status, emergency procedure, preoperative sepsis, and type of surgery (Table 4). None of the second-order interaction terms involving surgery was chosen, which suggests that there was not substantial interaction between surgery type and any of the other variables in the model.

Development and Validation of Risk Model

The 2007 data set was used as the training set in order to develop the model, and the 2008 data set served as the validation set. The risk model included significant predictors from the 2007 data set. The parameter estimates and their SEs are summarized in Table 4.

Table 4 can be used to generate probability estimates identical to the risk calculator by inserting the

appropriate coefficient estimates into the standard logistic regression model to compute the estimated logit and then translating this logit into the probability scale as described in the “Materials and Methods” section.

The c-statistic for the training set was 0.907 in the 21-variable model and 0.894 in the final model, indicating excellent discrimination. Figures 1 (21-variable model) and 2 (final model) show that the calibration (Hosmer-Lemeshow goodness-of-fit test) was excellent in both models, without a substantial deviation from the 45-degree line of perfect fit.

The selected risk model (final model) was then applied to the 2008 validation set. The c-statistic that arose from using the 2007 model to estimate PRF probability in the 2008 data was 0.897, indicating excellent discrimination. These findings indicate that the model performance was very similar in both the 2007 training set and the 2008 validation set, with the model continuing to have excellent discrimination in an independent data set.

Development of Risk Calculator

The selected model was then used to develop an interactive risk calculator. When the required input is entered into this calculator for a given patient, it returns a model-based percent estimate of PRF.

DISCUSSION

Over the past few decades, significant emphasis has been given to identification of risk factors for postoperative cardiac complications.^{17,20-22} In comparison, few studies have assessed risk factors for pulmonary complications, including PRF—one of the most serious pulmonary complications. Khuri et al²³

Table 3—Univariate Analysis—Perioperative (2007 NSQIP Data Set)

Variables	PRF (n = 6,531)	No PRF (n = 204,879)
Intraoperative/postoperative		
Intraoperative PRBC transfusion in units	1.6 ± 3.6	0.1 ± 0.8
Transfusion > 4 units PRBC within 72 h	7.92	0.26
Graft/flap failure with return to operating room	1.10	0.27
Cardiac arrest	9.71	0.13
Myocardial infarction	2.53	0.09
DVT	6.94	0.52
Pulmonary embolism	2.22	0.25
Pneumonia	33.59	0.80
Reintubation/unplanned intubation	47.97	0.00
Ventilator > 48 h	80.02	0.00
Renal insufficiency with rise in creatinine by 2 (no dialysis)	5.24	0.22
Renal failure requiring dialysis	10.29	0.21
Superficial site infection	6.78	2.97
Deep incisional infection	4.00	0.73
Organ/space infection	10.32	1.10
Urinary tract infection	9.60	1.45
Wound and fascia disruption	5.44	0.50
Coma	2.60	0.02
Stroke	3.02	0.16
Peripheral nerve injury	0.34	0.07
Septic shock	34.44	0.54
Sepsis	18.79	1.77
Operative time in min	174.9 ± 135.5	107.1 ± 86.1
Anesthesia time in min	248.4 ± 152.8	158.5 ± 100.6
Return to operating room	39.70	4.57
Total hospital length of stay, d	28.2 ± 26.5	3.9 ± 7.8
Morbidity (development of any of the above 22 postoperative complications)	100.00	8.79
Death within 30 d	25.62	0.98
Type of surgery		
Aorta	8.7	2.0
Cardiac	0.9	0.3
GBAAS	6.0	17.3
FG/HPB	14.8	5.5
Intestine	37.1	14.3
Brain	0.5	0.2
Orthopedic	3.3	4.4
Other abdomen	6.9	1.8
Skin	2.4	2.3
Nonesophageal thoracic	1.4	0.6
Urology	0.5	0.9
Peripheral vascular	10.4	8.2
Anorectal	0.3	1.6
Bariatric	1.3	5.9
Breast	0.2	10.4
ENT	0.1	0.3
OB/GYN	0.2	1.4
Hernia	4.2	15.3
Neck	0.8	4.9
Spine	0.2	0.7
Vein	0.1	1.6

All values except continuous variables are expressed as percentages. Continuous variables are expressed as mean ± SD. All *P* values ≤ .0001. See Table 1 and 2 legends for expansion of abbreviations.

found PRF to be an independent predictor of 30-day and long-term mortality. In their adjusted analysis, Dimick et al²⁴ found respiratory complications to be associated with the largest attributable cost as well as the largest increase in hospital length of stay (by 5.5 days). These studies collectively sug-

gest that pulmonary complications may be as deleterious to postoperative outcomes as cardiac and other complications.

This study is the first to attempt to study PRF across a broad population base that includes both genders, academic and community hospitals, and multiple

Table 4—Estimates, SEs, and Variables Associated With PRF in the Stepwise Logistic Regression Analysis (2007 NSQIP Data Set—Final Model)

Parameter	Estimate	SE	Model Term	Adjusted OR	95% Wald CI
Intercept	−1.7397	0.1504	β_0		
Totally dependent functional status ^a	1.4046	0.0519	β_1	4.07	3.68-4.51
Partially dependent functional status ^a	0.7678	0.0422	β_2	2.16	1.98-2.34
ASA class 1 ^b	−3.5265	0.2672	β_3	0.03	0.02-0.05
ASA class 2 ^b	−2.0008	0.1184	β_4	0.14	0.11-0.17
ASA class 3 ^b	−0.6201	0.1081	β_5	0.54	0.44-0.67
ASA class 4 ^b	0.2441	0.1053	β_6	1.28	1.04-1.57
Preoperative sepsis (none) ^c	−0.7840	0.0444	β_7	0.46	0.42-0.50
Preoperative sepsis ^c	0.2752	0.0645	β_8	1.32	1.16-1.49
Preoperative septic shock ^c	0.9035	0.0675	β_9	2.47	2.16-2.82
Emergency case (absence vs presence)	−0.5739	0.0378	β_{10}	0.56	0.52-0.61
Anorectal ^d	−1.3530	0.2710	β_{11}	0.26	0.15-0.44
Aortic ^d	1.0781	0.1149	β_{12}	2.94	2.35-3.68
Bariatric ^d	−1.0112	0.1495	β_{13}	0.36	0.27-0.49
Brain ^d	0.7336	0.3043	β_{14}	2.08	1.15-3.78
Breast ^d	−2.6462	0.2706	β_{15}	0.07	0.04-0.12
Cardiac ^d	0.2744	0.1811	β_{16}	1.32	0.92-1.88
ENT ^d	0.1060	0.7368	β_{17}	1.11	0.26-4.71
Foregut/hepatopancreatobiliary ^d	0.9694	0.1094	β_{18}	2.64	2.13-3.27
GBAAS ^d	−0.5668	0.1166	β_{19}	0.57	0.45-0.71
Intestinal ^d	0.5737	0.1057	β_{20}	1.78	1.44-2.18
Neck ^d	−0.5271	0.3023	β_{21}	0.59	0.33-1.07
OB/GYN ^d	−1.2431	0.6004	β_{22}	0.29	0.09-0.94
Orthopedic ^d	−0.8577	0.1286	β_{23}	0.42	0.33-0.55
Other abdomen ^d	0.2416	0.1228	β_{24}	1.27	1.001-1.62
Peripheral vascular ^d	−0.2389	0.1111	β_{25}	0.79	0.63-0.98
Skin ^d	−0.3206	0.1396	β_{26}	0.73	0.55-0.95
Spine ^d	−0.5220	0.4337	β_{27}	0.593	0.25-1.39
Thoracic ^d	0.6715	0.1595	β_{28}	1.96	1.43-2.68
Vein ^d	−2.0080	0.5128	β_{29}	0.134	0.05-0.37
Urology ^d	0.3093	0.2616	β_{30}	1.36	0.82-2.28

The estimate and the SE refer to the estimate of the logistic regression coefficient for the specific variable and its associated SE. C-statistic, 0.894.

See Table 1 and 2 legends for expansion of abbreviations.

^aReference group, independent functional status.

^bReference group, ASA class 5.

^cReference group, preoperative systemic inflammatory response syndrome.

^dReference group, hernia surgery.

surgical subspecialties. PRF was seen in 3.1% of patients in this study, which compares to 3.4% and 3.0% as reported in the two prior (2000 and 2007) VASQIP studies.^{7,8} Death within 30 days was seen in 26% of patients with PRF. This again compares well to the VASQIP studies, where 30-day mortality rates of 27% were reported. Rates of PRF and the associated 30-day mortality are thus strikingly similar between the academic, community, and VA hospitals across the country. Our observed mortality rate also suggests that there has been minimal change in the incidence of PRF or its attendant mortality over the last 10 years.

The impact of performance status on outcomes is well documented in literature, with many studies reporting that patients with limited independence have poor outcomes.²⁵⁻²⁷ Like Arozullah et al,⁷ this study also found dependent functional status to be a risk factor associated with PRF. Higher ASA class,

emergency case, and preoperative sepsis were also found to be associated with PRF. Overall, the type of surgery performed had the largest difference in terms of risk for PRF, with brain, foregut/hepatopancreatobiliary, and aortic surgeries being associated with the highest risk.

The PRF risk calculator was developed to aid in the surgical decision making and informed consent process. It is in the form of an interactive spreadsheet and is available online at <http://www.surgicalriskcalculator.com/prf-risk-calculator> for free download.

A few previous studies in other disciplines have used logistic regression models to create point-based score systems. We instead chose to develop a risk calculator based directly on the logistic regression model. This approach allowed direct modeling and prediction of PRF, rather than using one model to assess PRF and another to predict risk based on a point system. Hence, no loss of accuracy for using a

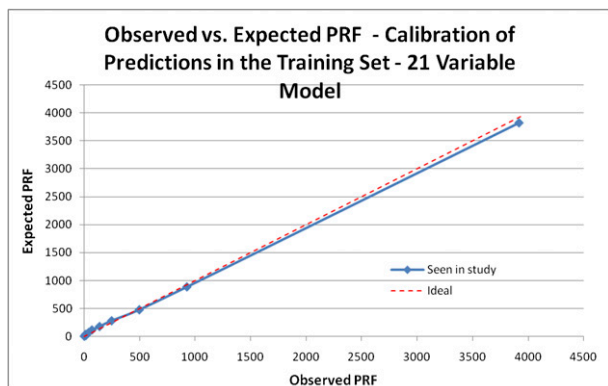


FIGURE 1. Calibration of predictions of PRF in the training set (21-variable model). ♦ Denotes deciles of patients. PRF = postoperative respiratory failure.

second model is incurred with this strategy. In addition, the previous indices did not provide an actuarial estimate of risk; instead, patients were classified as being at high risk/intermediate risk/low risk for developing complications. With the PRF risk calculator, we instead provide an exact model-based estimate of PRF probability for a patient. This approach is more precise than a point system, but it may be less simple for some users to implement. However, as clinicians take advantage of new, hand-held computer-based technologies to use pharmacopeias and clinical management guidelines, it is our belief that a risk calculator will find widespread use and assist physicians and surgeons in making clinical decisions.

Apart from identifying high-risk patients, we foresee the risk calculator as an important tool in the informed consent process. The process of patient-centered informed consent requires the presentation of adequate information regarding risks and benefits.²⁸ Accurate individualized assessment of PRF, which contributes greatly to morbidity and mortality, would certainly assist in meeting the latter objective.

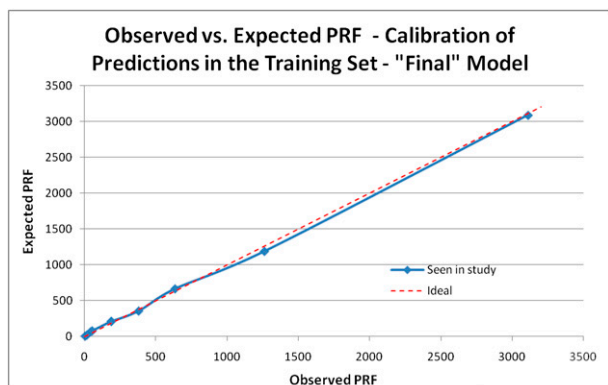


FIGURE 2. Calibration of predictions of PRF in the training set (final model). ♦ Denotes deciles of patients. See Figure 1 legend for expansion of abbreviation.

Physicians have long quoted the most current literature to explain risks of adverse outcomes associated with a procedure. This has not always been an easy task, as each patient is different with a unique set of risk factors. Thus, this risk calculator will simplify the informed consent process by estimating the risk of PRF, and we envision its use preoperatively by the physicians/surgeons. It may also be used to hold ICU beds prior to surgery for patients who otherwise appear to be at low risk.

In spite of its many strengths, this study has a few limitations. Variables analyzed were limited to those recorded by NSQIP. Despite the data set being fairly comprehensive, with >50 preoperative variables, some comorbidities, such as obstructive sleep apnea and history of venous thromboembolism, were not included. Similarly, pulmonary function test results may be relevant to many of the comorbidities and surgeries but are not available in NSQIP. Information on hospital volume is also not contained in NSQIP. The results of this study may not apply to hospitals that are not a part of NSQIP. However, this is unlikely given its diversity. Finally, whereas data collection is prospective in NSQIP, these data were retrospectively analyzed for the development and validation of the risk calculator.

In conclusion, PRF occurs postoperatively in around 3% of patients, and 25% of patients with PRF die within 30 days. PRF incidence is similar in academic, community, and VA hospitals, with similar effects on 30-day mortality. There has been no decline in these numbers in the last decade. The high association of PRF with mortality emphasizes the importance of risk estimation and preoperative optimization. This risk calculator, with its high discriminative/predictive ability for PRF, is a step in that direction.

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